
Invariants

UTSA Problem Solving Club • Fall 2026

Breaking the Bar

You have a rectangular bar of chocolate made of $m \times n$ small squares. A *break* snaps one piece in a straight line along the grooves, into two pieces.

- (a) How many breaks does it take to reduce the bar to its mn separate squares?
- (b) Two players take turns breaking, and whoever makes the last break wins. Which player should win, and for which m and n ?

The Chameleons

On an island live 13 red, 15 green and 17 blue chameleons. Whenever two chameleons of *different* colors meet, both change into the third color. (So a red and a green meeting leaves two more blues and one fewer of each of the others.)

Can the island ever reach a state in which every chameleon is the same color?

A Bean in the Can

A can holds some white beans and some black beans. Repeat the following until one bean is left: take out two beans at random; if they are the same color, drop a *black* bean in; if they differ, drop a *white* bean in. (Assume you have a bag of spare beans to draw replacements from.)

What is the color of the final bean?

The Blackboard

The numbers $1, 2, 3, \dots, n$ are written on a blackboard. Repeatedly erase any two of them, say a and b , and write the single number $a + b - 1$ in their place. Continue until one number remains.

What are the possible values of the last number remaining?

A Knight from Corner to Corner

A knight starts in one corner of an ordinary 8×8 chessboard. It wants to visit every one of the 64 squares exactly once and finish on the diagonally opposite corner.

Can it be done? If so, show how.

Coins in a Line

An even number of coins lie in a row, of distinct values. Two players take turns removing a coin from either end of the row, keeping it.

Show that the first player can always finish with at least as much money as the second.

Going Further

He Loves Me, He Loves Me Not

A daisy has n petals arranged in a circle. Two players take turns removing either a single petal, or two adjacent petals. Whoever removes the last petal wins.

For which n does the first player have a winning strategy?

The Fifteen Puzzle

Fifteen numbered tiles sit in a 4×4 frame, leaving one empty square; a move slides a tile into the empty square. Start from the solved position, then lift out the tiles numbered 14 and 15 and put them back swapped.

Show that no sequence of moves returns the puzzle to the solved position.

Red Squares, Black Squares

Let n be an even positive integer. Write the numbers $1, 2, \dots, n^2$ in the squares of an $n \times n$ grid so that the k -th row, from left to right, is

$$(k-1)n+1, (k-1)n+2, \dots, (k-1)n+n.$$

Color the squares of the grid so that half of the squares in each row and in each column are red and the other half are black (a checkerboard coloring is one possibility). Prove that for each coloring, the sum of the numbers on the red squares is equal to the sum of the numbers on the black squares.

Source: Putnam 2001 B1

Pegs in a Row

Let k and n be integers with $1 \leq k < n$. Alice and Bob play a game with k pegs in a line of n holes. At the beginning of the game, the pegs occupy the k leftmost holes. A legal move consists of moving a single peg to any vacant hole that is further to the right. The players alternate moves, with Alice playing first. The game ends when the pegs are in the k rightmost holes, so whoever is next to play cannot move and therefore loses. For what values of n and k does Alice have a winning strategy?

Source: Putnam 2020 B2